

05 Weighted Rolling Validation (Toy Example)

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Simulation process and logic

This section describes the full simulation pipeline: what is being compared, how one-step rolling validation is scored, how power weights enter, and how Monte Carlo estimates are formed.

What the simulation is asking

We have two predictors for a sequence $Y_1, \dots, Y_n$:

- **Model 0:** at time i , predict with the running sample mean of the past,

$$\hat{\mu}_{i-1} = \frac{1}{i-1} \sum_{j=1}^{i-1} Y_j.$$

- **Model 1 (oracle):** always predict 0.

The true mean is 0, so Model 1 is correct. Rolling validation builds a score $\Xi_n^{(w)}$. We incorrectly select Model 0 when $\Xi_n^{(w)} < 0$. The target quantity is

$$P(\Xi_n^{(w)} < 0)$$

as a function of sample size n and weight exponent α .

Indices: n , k , and i

- n : total length of one data path $Y_1, \dots, Y_n$.
- i : validation time, $i = 2, \dots, n$. At time i we predict Y_i using only the past.
- $k = i - 1$: training size (number of past observations used to form $\hat{\mu}_{i-1}$).

So each one-step task is: train on $Y_1, \dots, Y_k$, validate on $Y_{k+1} = Y_i$.

One-step rolling validation (unweighted increment)

For each $i = 2, \dots, n$, squared-error losses are

$$L_0 = (Y_i - \hat{\mu}_{i-1})^2, \quad L_1 = (Y_i - 0)^2 = Y_i^2.$$

The loss difference simplifies to the **unweighted increment**

$$L_0 - L_1 = -2Y_i\hat{\mu}_{i-1} + \hat{\mu}_{i-1}^2.$$

In code this is computed for an entire path at once (via cumulative means), returning a length- $(n-1)$ vector of increments for $i = 2, \dots, n$. That vector **is** the expanding one-step RV process; weights and summation come next.

Power weights and the score Ξ

A power weight at validation time i is

$$w_i = i^\alpha.$$

For a fixed α and n , the weight vector is

$$(2^\alpha, 3^\alpha, \dots, n^\alpha),$$

one weight per validation time (not a single weight n^α).

The weighted rolling-validation score is the weighted sum of increments:

$$\Xi_n^{(w)} = \sum_{i=2}^n w_i (-2Y_i\hat{\mu}_{i-1} + \hat{\mu}_{i-1}^2) = \sum_{i=2}^n i^\alpha (-2Y_i\hat{\mu}_{i-1} + \hat{\mu}_{i-1}^2).$$

Special case $\alpha = 0$: every $w_i = 1$, so Ξ_n is the classical unweighted sum (the **control**).

Sign convention:

- $\Xi_n^{(w)} > 0$: prefers Model 1 (oracle),
- $\Xi_n^{(w)} < 0$: prefers Model 0 (incorrect selection in this toy).

Theoretical mean (used as a check)

Because $Y_i \sim N(0, 1)$ is independent of $\hat{\mu}_{i-1}$ and $E[Y_i] = 0$,

$$E[-2Y_i\hat{\mu}_{i-1}] = 0, \quad E[\hat{\mu}_{i-1}^2] = \frac{1}{i-1}.$$

Hence

$$E[\Xi_n^{(w)}] = \sum_{i=2}^n \frac{w_i}{i-1} = \sum_{i=2}^n \frac{i^\alpha}{i-1}.$$

The simulation reports both the Monte Carlo mean of Ξ and this exact theoretical mean.

Monte Carlo grid: how one experiment is run

Data are $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(0, 1)$. Random seed is fixed at 2026 inside the grid runner.

For a chosen list of sample sizes and alphas, and `mc_runs` replications, the efficient procedure is:

1. Let $n_{\max}$ be the largest sample size on the grid.
2. **Precompute** weight vectors $w_i = i^\alpha$ for every pair (α, n) .
3. **Precompute** theoretical means $\sum_{i=2}^n w_i / (i - 1)$ for every pair (α, n) .
4. For each Monte Carlo replication $r = 1, \dots, \text{mc_runs}$:
 - draw one long path $Y_1, \dots, Y_{n_{\max}}$;
 - compute the full unweighted increment vector for $i = 2, \dots, n_{\max}$ **once**;
 - for each α and each n on the grid:
 - take the first $n - 1$ increments (as if the path had length n);
 - form $\Xi = \sum_{i=2}^n w_i \cdot \text{inc}_i$;
 - store that value as replication r for (α, n) .
5. For each (α, n) , summarize the `mc_runs` stored values:
 - $\hat{P}(\Xi < 0)$ (incorrect-selection rate),
 - sample mean and variance of Ξ ,
 - theoretical mean,
 - Chebyshev ratio $R_n = \widehat{\text{Var}}(\Xi) / (\widehat{\text{mean}})^2$.

Reusing one long path across all smaller n and all alphas makes the grid much cheaper than resimulating a new path for every (α, n) .

Two experiments in this report

- **Broad grid:** $\alpha \in \{-1, -0.5, 0, 0.5, 1\}$, with $\alpha = 0$ as the unweighted control.
- **Near-zero grid:** $\alpha \in \{-0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5\}$, same n -grid and Monte Carlo size, to look more carefully around the control.

Throughout the report we distinguish:

- **mathematical derivation** (exact or asymptotic formulas),
- **Monte Carlo evidence** (simulated means, variances, error rates),
- **interpretation** (informal reading of plots; not a theorem).

Goal

This report studies a **Gaussian toy example** of rolling-validation consistency and asks whether the incorrect-selection probability still goes to zero after **power weights** $w_i = i^\alpha$. We keep $\alpha = 0$ as a control, compare a broad set of power weights, then run a second experiment focused on alphas near zero.

Toy-example setup (formal summary)

- Data: $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(0, 1)$.
- Model 0: running mean $\hat{\mu}_{i-1}$; Model 1: predict 0.
- Score:

$$\Xi_n^{(w)} = \sum_{i=2}^n i^\alpha (-2Y_i \hat{\mu}_{i-1} + \hat{\mu}_{i-1}^2).$$

- Incorrect selection when $\Xi_n^{(w)} < 0$.

Table 1: Simulation settings for analysis 05.

setting	value
seed	2026
Monte Carlo replications	1000
Sample sizes (grid)	100, 200, 500, 1000, 2000, 5000, 10000
weight family	$w_i = i^\alpha$
broad alphas (0 = control first)	0, -1, -0.5, 0.5, 1
near-zero alphas	-0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5
incorrect selection	$\Xi_n < 0$

Original control: $\alpha = 0$

For the unweighted case, theory gives $E[\Xi_n] \sim \log n$ and $\text{Var}(\Xi_n) = O(1)$, so $P(\Xi_n < 0) \rightarrow 0$.

Table 2: Control results under alpha = 0 (Monte Carlo evidence).

n	p_neg	mean_xi	theory_mean	log_n	var_xi	R
100	0.027	5.2627	5.1774	4.6052	9.9932	0.3608
200	0.017	5.9071	5.8730	5.2983	9.9661	0.2856
500	0.008	6.9230	6.7908	6.2146	9.4043	0.1962
1000	0.006	7.5184	7.4845	6.9078	9.6778	0.1712

n	p_neg	mean_xi	theory_mean	log_n	var_xi	R
2000	0.003	8.2355	8.1779	7.6009	10.2321	0.1509
5000	0.001	9.2193	9.0943	8.5172	9.3939	0.1105
10000	0.001	9.8940	9.7875	9.2103	9.4289	0.0963

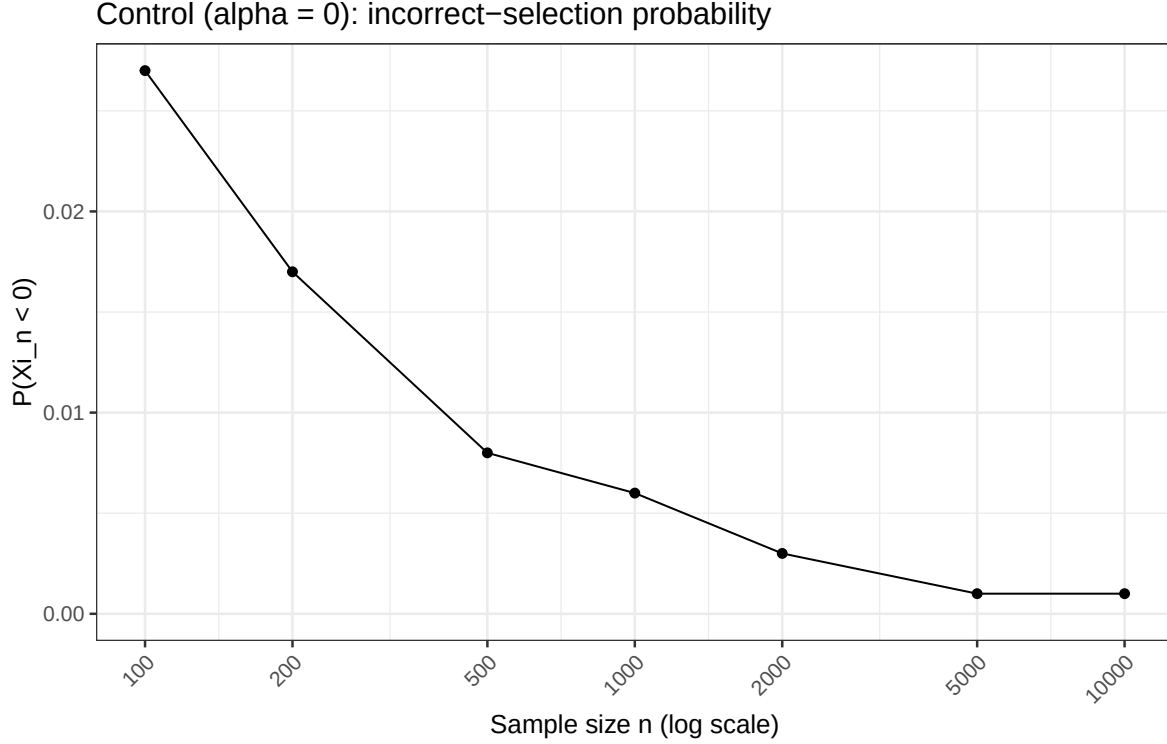


Figure 1: Control (alpha = 0): incorrect-selection probability vs n.

Broad power-weight comparison

We compare $\alpha \in \{-1, -0.5, 0, 0.5, 1\}$ with the same n -grid and 1000 Monte Carlo replications. Plots use a log scale on n , with a tick at every sample size, and include 0 on the y -axis (except the variance plot, which uses a log y -scale so 0 cannot appear).

Correct theoretical mean derivation

Derivation. Expanding the one-step difference and using independence of Y_i and $\hat{\mu}_{i-1}$ with $E[Y_i] = 0$ gives $E[-2Y_i\hat{\mu}_{i-1}] = 0$ and $E[\hat{\mu}_{i-1}^2] = 1/(i-1)$. Therefore

$$E[\Xi_n^{(w)}] = \sum_{i=2}^n \frac{i^\alpha}{i-1}.$$

Since $i^\alpha/(i-1) \approx i^{\alpha-1}$, the expected signal has order:

- $\alpha = -1$: $E[\Xi_n^{(w)}] = O(1)$
- $\alpha = -0.5$: $E[\Xi_n^{(w)}] = O(1)$
- $\alpha = 0$: $E[\Xi_n^{(w)}] \sim \log n$
- $\alpha = 0.5$: $E[\Xi_n^{(w)}] \asymp 2\sqrt{n}$
- $\alpha = 1$: $E[\Xi_n^{(w)}] \asymp n$

A diverging mean is **not** enough for consistency via Chebyshev; one also needs the ratio R_n to go to zero. Variance and R_n behavior below are **Monte Carlo evidence**, not formal theorems.

Normalized weights. Multiplying all weights by a common positive constant (for example replacing i^α by $(i/n)^\alpha$) does not change the sign of $\Xi_n^{(w)}$, so it does not change the selection event.

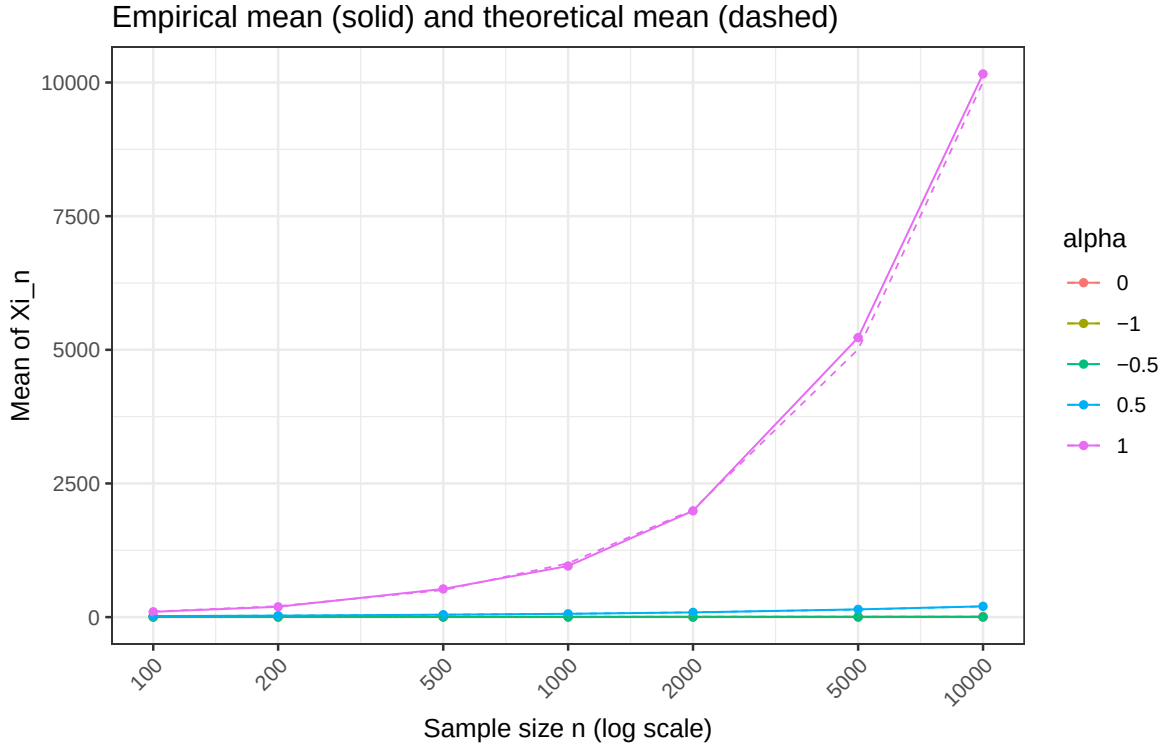


Figure 2: Empirical mean (solid) and theoretical mean (dashed) of Ξ_n versus n .

Incorrect-selection probability results

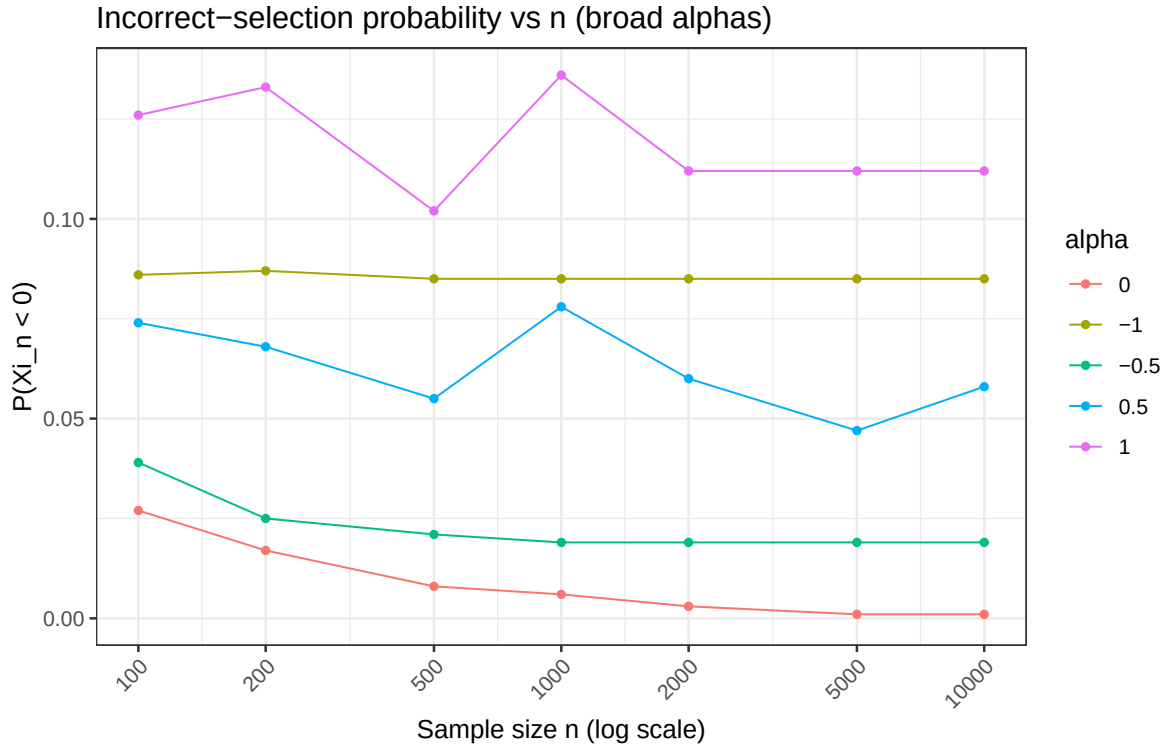


Figure 3: $P(X_{i_n} < 0)$ versus n for each broad power weight α .

The x -axis is log-scaled with a tick at every sample size on the grid. Linear y -axes include 0.

Variance results

Variance for $\alpha = 1$ is much larger than for the other alphas, so the y -axis uses a log scale so that $\alpha \in \{-1, -0.5, 0, 0.5\}$ remain visible.

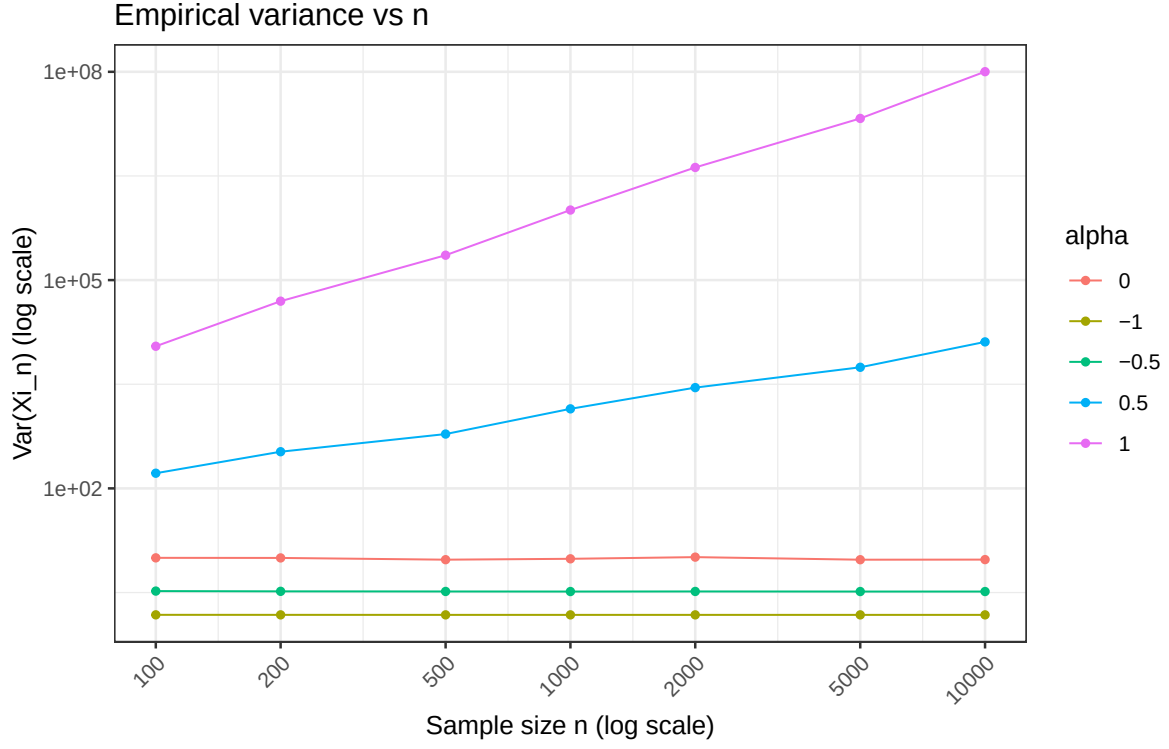


Figure 4: Empirical variance of Xi_n versus n (log y-scale).

Chebyshev ratio results

$$R_n = \frac{\text{Var}(\Xi_n^{(w)})}{(E[\Xi_n^{(w)}])^2}.$$

If $R_n \rightarrow 0$, Chebyshev would imply $P(\Xi_n^{(w)} < 0) \rightarrow 0$. A single small value of R_n at one n does **not** prove consistency. We look at the **trend** in n .

For $\alpha = 0$, R_n clearly decreases with n (Monte Carlo evidence). For the other broad alphas in this experiment, the simulation does not show the same clear tendency toward zero. This is simulation evidence only.

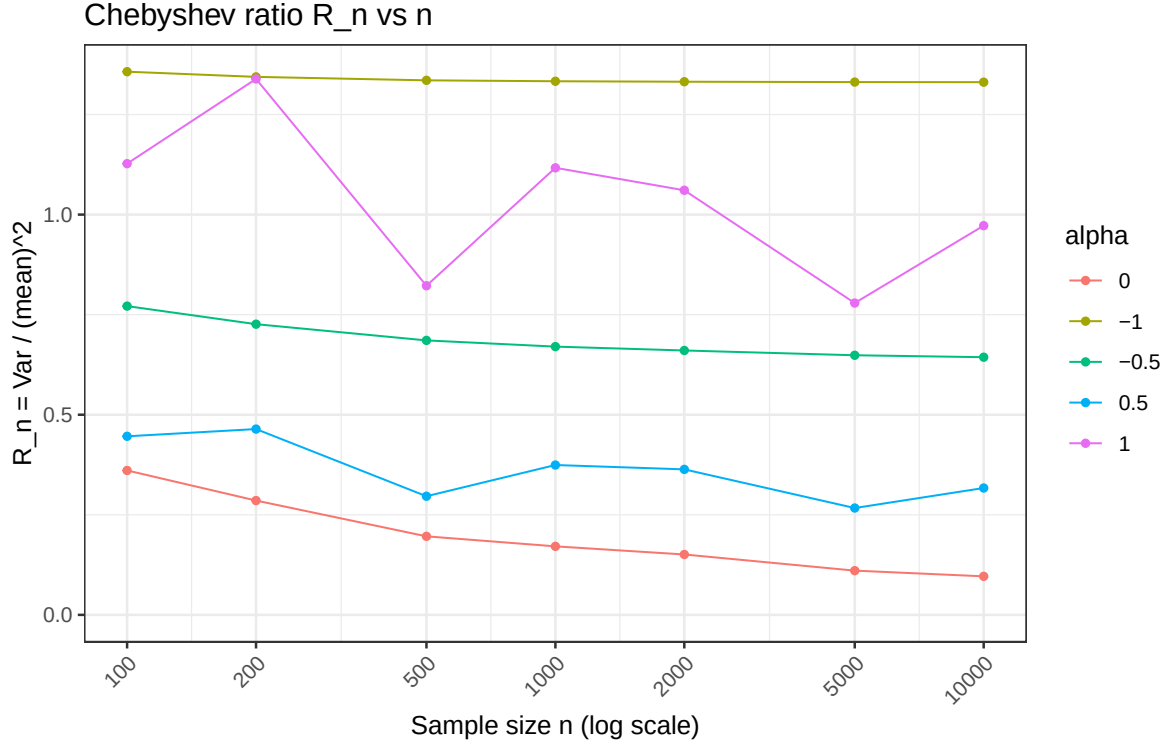


Figure 5: Chebyshev ratio R_n versus n (Monte Carlo evidence).

Near-zero-alpha experiment

Because the broad grid suggests that $\alpha = 0$ behaves differently from more strongly weighted cases, we run a second experiment with

$$\alpha \in \{-0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5\},$$

using the same `mc_runs` = 1000 and the same n -grid. This plot is separate from the broad-alpha comparison.

Table 3: Settings for the near-zero-alpha experiment.

setting	value
seed	2026
Monte Carlo replications	1000
Sample sizes (grid)	100, 200, 500, 1000, 2000, 5000, 10000

setting	value
alphas (near zero)	-0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5

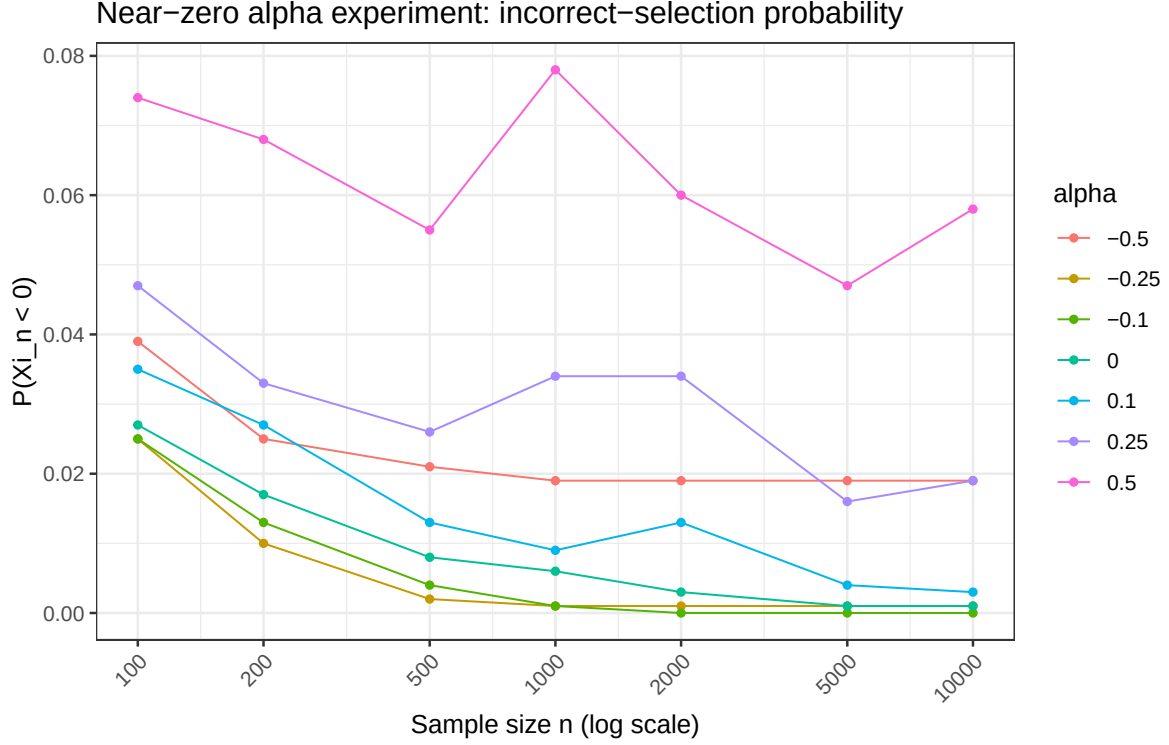


Figure 6: Near-zero alpha experiment: $P(Xi_n < 0)$ versus n .

Table 4: Near-zero alpha Monte Carlo results.

alpha	n	p_neg	mean_xi	theory_mean	var_xi	R
-0.50	100	0.039	2.0742	1.9838	3.3187	0.7714
-0.50	200	0.025	2.1286	2.0425	3.2905	0.7262
-0.50	500	0.021	2.1862	2.0946	3.2772	0.6857
-0.50	1000	0.019	2.2087	2.1208	3.2693	0.6702
-0.50	2000	0.019	2.2279	2.1393	3.2783	0.6605
-0.50	5000	0.019	2.2455	2.1557	3.2701	0.6486
-0.50	10000	0.019	2.2535	2.1640	3.2690	0.6437
-0.25	100	0.025	3.1727	3.0742	5.2768	0.5242
-0.25	200	0.010	3.3596	3.2760	5.1647	0.4576
-0.25	500	0.002	3.6009	3.4940	5.0616	0.3904

alpha	n	p_neg	mean_xi	theory_mean	var_xi	R
-0.25	1000	0.001	3.7166	3.6287	5.0216	0.3635
-0.25	2000	0.001	3.8337	3.7419	5.0748	0.3453
-0.25	5000	0.001	3.9650	3.8644	4.9920	0.3175
-0.25	10000	0.001	4.0384	3.9401	4.9746	0.3050
-0.10	100	0.025	4.2524	4.1572	7.4700	0.4131
-0.10	200	0.013	4.6451	4.5811	7.3013	0.3384
-0.10	500	0.004	5.2164	5.0973	6.9870	0.2568
-0.10	1000	0.001	5.5254	5.4573	6.9599	0.2280
-0.10	2000	0.000	5.8727	5.7930	7.1397	0.2070
-0.10	5000	0.000	6.3121	6.2025	6.7907	0.1704
-0.10	10000	0.000	6.5898	6.4883	6.7271	0.1549
0.00	100	0.027	5.2627	5.1774	9.9932	0.3608
0.00	200	0.017	5.9071	5.8730	9.9661	0.2856
0.00	500	0.008	6.9230	6.7908	9.4043	0.1962
0.00	1000	0.006	7.5184	7.4845	9.6778	0.1712
0.00	2000	0.003	8.2355	8.1779	10.2321	0.1509
0.00	5000	0.001	9.2193	9.0943	9.3939	0.1105
0.00	10000	0.001	9.8940	9.7875	9.4289	0.0963
0.10	100	0.035	6.6127	6.5478	14.3974	0.3293
0.10	200	0.027	7.6705	7.6898	15.2461	0.2591
0.10	500	0.013	9.4782	9.3227	14.6160	0.1627
0.10	1000	0.009	10.6258	10.6599	16.6020	0.1470
0.10	2000	0.013	12.1070	12.0926	18.9244	0.1291
0.10	5000	0.004	14.3118	14.1450	17.5529	0.0857
0.10	10000	0.003	15.9514	15.8272	19.4860	0.0766
0.25	100	0.047	9.5905	9.5911	29.9546	0.3257
0.25	200	0.033	11.8172	11.9951	38.5119	0.2758
0.25	500	0.026	16.1142	15.8754	42.9997	0.1656
0.25	1000	0.034	19.1872	19.4574	64.5100	0.1752
0.25	2000	0.034	23.5875	23.7152	90.2383	0.1622
0.25	5000	0.016	30.9934	30.6029	105.4870	0.1098
0.25	10000	0.019	37.2094	36.9676	160.2820	0.1158
0.50	100	0.074	19.2532	19.5734	165.3721	0.4461
0.50	200	0.068	26.9661	27.9018	337.5298	0.4642
0.50	500	0.055	45.2258	44.3779	606.1987	0.2964
0.50	1000	0.078	61.1276	62.9218	1398.7535	0.3743
0.50	2000	0.060	88.1974	89.1328	2827.6184	0.3635
0.50	5000	0.047	144.1827	141.1238	5549.3425	0.2669
0.50	10000	0.058	201.5904	199.7087	12878.9444	0.3169

Table 5: Near-zero alpha summary (derivation vs simulation interpretation).

Alpha	Weight	Expected signal	Error probability	R behavior	Interpretation
-0.50	$i^{-0.5}$	bounded $O(1)$ (derivation)	decreases then levels near 0.02 (simulation)	slow decrease; stays large (simulation)	levels off away from 0 (simulation interpretation)
-0.25	$i^{-0.25}$	bounded $O(1)$ (derivation)	decreases toward ~ 0 (simulation)	decreases with n (simulation)	looks similar to control at these n (simulation interpretation)
-0.10	$i^{-0.1}$	bounded $O(1)$ (derivation)	decreases to ~ 0 (simulation)	decreases with n (simulation)	looks similar to control (simulation interpretation)
0.00	1	$\sim \log(n)$ (derivation)	decreases toward 0 (simulation)	decreases with n (simulation)	clear evidence toward 0 (simulation interpretation)
0.10	$i^{0.1}$	$\sim c n^{0.1}$ (derivation)	decreases toward ~ 0 (simulation)	decreases with n (simulation)	looks similar to control (simulation interpretation)
0.25	$i^{0.25}$	$\sim c n^{0.25}$ (derivation)	decreases then stays noisy near 0.02 (simulation)	mild decrease; noisy (simulation)	weaker / unclear vs $\alpha = 0$ (simulation interpretation)
0.50	$\sqrt{i}$	$\sim 2 \sqrt{n}$ (derivation)	noisy and remains positive (simulation)	no clear tendency toward 0 (simulation)	no clear convergence to 0 (simulation interpretation)

Summary and interpretation

The tables below mix **derivation** (expected signal order) with **Monte Carlo** descriptions and **simulation interpretations**. They do not claim a theorem for weighted cases. The first covers the broad alpha grid; the near-zero summary appears above with that experiment.

Table 6: Summary by alpha (derivation vs simulation interpretation).

Alpha	Weight	Expected signal	Error probability	R behavior	Interpretation
-1.0	i^{-1}	bounded $O(1)$ (derivation)	approximately flat near 0.08 (simulation)	no clear tendency toward 0 (simulation)	appears bounded away from 0 (simulation interpretation)
-0.5	$i^{-0.5}$	bounded $O(1)$ (derivation)	decreases then levels near 0.01 (simulation)	no clear tendency toward 0 (simulation)	decreases initially, then appears to level off (simulation interpretation)
0.0	1	$\sim \log(n)$ (derivation)	decreases toward 0 (simulation)	decreases with n (simulation)	clear evidence toward 0 (simulation interpretation)
0.5	$\sqrt{i}$	$\sim 2 \sqrt{n}$ (derivation)	noisy and remains positive (simulation)	no clear tendency toward 0 (simulation)	unclear; no clear convergence to 0 (simulation interpretation)
1.0	i	$\sim n$ (derivation)	remains around 0.1 (simulation)	no clear tendency toward 0 (simulation)	appears bounded away from 0 (simulation interpretation)

Full numeric results (broad grid)

Table 7: Full Monte Carlo results by alpha and n (broad grid).

alpha	n	p_neg	mean_xi	theory_mean	var_xi	R
-1.0	100	0.086	1.0547	0.9900	1.509600e+00	1.3570
-1.0	200	0.087	1.0594	0.9950	1.508300e+00	1.3440
-1.0	500	0.085	1.0627	0.9980	1.508100e+00	1.3354
-1.0	1000	0.085	1.0635	0.9990	1.507900e+00	1.3331
-1.0	2000	0.085	1.0641	0.9995	1.508100e+00	1.3320
-1.0	5000	0.085	1.0644	0.9998	1.508000e+00	1.3311
-1.0	10000	0.085	1.0645	0.9999	1.507900e+00	1.3308
-0.5	100	0.039	2.0742	1.9838	3.318700e+00	0.7714
-0.5	200	0.025	2.1286	2.0425	3.290500e+00	0.7262
-0.5	500	0.021	2.1862	2.0946	3.277200e+00	0.6857
-0.5	1000	0.019	2.2087	2.1208	3.269300e+00	0.6702
-0.5	2000	0.019	2.2279	2.1393	3.278300e+00	0.6605
-0.5	5000	0.019	2.2455	2.1557	3.270100e+00	0.6486
-0.5	10000	0.019	2.2535	2.1640	3.269000e+00	0.6437
0.0	100	0.027	5.2627	5.1774	9.993200e+00	0.3608
0.0	200	0.017	5.9071	5.8730	9.966100e+00	0.2856
0.0	500	0.008	6.9230	6.7908	9.404300e+00	0.1962
0.0	1000	0.006	7.5184	7.4845	9.677800e+00	0.1712
0.0	2000	0.003	8.2355	8.1779	1.023210e+01	0.1509
0.0	5000	0.001	9.2193	9.0943	9.393900e+00	0.1105
0.0	10000	0.001	9.8940	9.7875	9.428900e+00	0.0963
0.5	100	0.074	19.2532	19.5734	1.653721e+02	0.4461
0.5	200	0.068	26.9661	27.9018	3.375298e+02	0.4642
0.5	500	0.055	45.2258	44.3779	6.061987e+02	0.2964
0.5	1000	0.078	61.1276	62.9218	1.398754e+03	0.3743
0.5	2000	0.060	88.1974	89.1328	2.827618e+03	0.3635
0.5	5000	0.047	144.1827	141.1238	5.549342e+03	0.2669
0.5	10000	0.058	201.5904	199.7087	1.287894e+04	0.3169
1.0	100	0.126	99.4276	104.1774	1.114409e+04	1.1273
1.0	200	0.133	192.6428	204.8730	4.968898e+04	1.3389
1.0	500	0.102	526.8505	505.7908	2.282400e+05	0.8223
1.0	1000	0.136	956.0595	1006.4845	1.020876e+06	1.1169
1.0	2000	0.112	1987.9989	2007.1779	4.192109e+06	1.0607
1.0	5000	0.112	5228.2595	5008.0943	2.129499e+07	0.7790
1.0	10000	0.112	10160.9230	10008.7875	1.003918e+08	0.9724